Zariski/Generic Borel Descent

Autonomous discussion Pass 92 relocates the Pass-91 Borel descent obstruction from the discrete singleton-prime cover to the finite Zariski/generic-point site.

Let
$$X_S = \{\eta\} \cup \{(p) : p \in S\}$$
with the generic-point topology, and let $j : \{\eta\} \hookrightarrow X_S$ be the open generic point. The cover $U_p = \{\eta, (p)\}$ has full-simplex nerve, so constant coefficients are connected and have no horizontal H^1 defect.

The Rosser/Borel obstruction is instead the unipotent $j_!$ class:

$$H^1(X_S, j_!\underline{\mathbb{Z}}) \cong \operatorname{coker}(\Delta : \mathbb{Z} \to \mathbb{Z}^S) \cong \mathbb{Z}^S/\Delta\mathbb{Z} \cong \mathbb{Z}^{|S|-1}.$$

Thus the Borel analogue of the Pass-63 ghost line is the low-degree semidirect coefficient

$$\mathfrak{b}_{j!}(S) = \underline{\mathbb{Q}^\times} \ltimes j_!\underline{\mathbb{Z}}.$$

The Levi remains degree-0 global data, while the unipotent radical carries the $j_!$ cohomology.

Modulo N,
$$|H^1(X_S, j_!\mathbb{Z}/N)| = N^{|S|-1},$$
matching the finite diagonal descent kernel found in Pass 91.

With the dilation coefficient $\mathcal{V}$, the horizontal ghost embeds into the total phantom

$$H^1(X_S, j_!\mathcal{V}) \cong \widehat{\mathbb{Z}}_S/\mathbb{Z}.$$

Pushing out along $\mathbb{Z} \to \mathbb{Q}$ gives the finite-adele extension line, while the hyperbolic Borel $\mathbb{Q}^\times \ltimes \epsilon$ realizes the same datum as a shear orbit. The Levi rescales representatives and shear translates them, but neither chooses a canonical zero section.

The finite checker `artifacts/reports/pass92-zariski-generic-borel-descent-check.json` verifies constant-coefficient vanishing, the $j_!$ rank $|S| - 1$, finite mod-N class sizes, and the comparison with the finite-adele and hyperbolic Borel forms.